

❖ HANDS-ON 3

❖ Directives & Pipes — Built-in and Custom

Task 1: Structural Directives

Goal: Control DOM structure dynamically using `*ngIf`, `*ngFor`, and `*ngSwitch`.

```
<> course-card.html U X TS course-list.ts U TS course-card.ts U <> course-list.html U <> home.html U
student-course-portal > src > app > components > course-card > <> course-card.html > ...
1 <h3>{{ course.name }}</h3>
2
3 <p><strong>ID:</strong> {{ course.id }}</p>
4
5 <p><strong>Code:</strong> {{ course.code }}</p>
6
7 <p><strong>Credits:</strong> {{ course.credits }}</p>
8
9 <div [ngSwitch]="course.gradeStatus">
10
11 <p *ngSwitchCase="'passed'" style="color: green;">
12 | Passed
13 </p>
14
15 <p *ngSwitchCase="'failed'" style="color: red;">
16 | Failed
17 </p>
18
19 <p *ngSwitchDefault style="color: gray;">
20 | Pending
21 </p>
22
23 </div>
24
25 <button (click)="enrollRequested.emit(course.id)">
26 | Enroll
27 </button>
28
29 </h3>
```

```
<> course-list.html U X <> course-card.html U # course-card.c: > > > > >
student-course-portal > src > app > pages > course-list > <> course-list.html > ...
1 <div *ngIf="isLoading">
2 | <p>Loading courses...</p>
3 </div>
4
5 <div *ngIf="!isLoading">
6
7 <div *ngIf="courses.length > 0; else noCourses">
8
9 <app-course-card
10 | *ngFor="let c of courses; let i = index; trackBy: trackBy(courseId)"
11 | [course]="c"
12 | (enrollRequested)="onEnroll($event)">
13 </app-course-card>
14
15 <p *ngIf="selectedCourseId">
16 | Selected course ID: {{ selectedCourseId }}
17 </p>
18
19 </div>
20
21 </div>
22
23 <ng-template #noCourses>
24 | <p>No courses available.</p>
25 </ng-template>
26
```

Student Course Portal

[Home](#)
[Courses](#)
[Profile](#)

Java Programming

ID: 1

Code: CS101

Credits: 4

Passed

Enroll

Angular

ID: 2

Code: CS201

Credits: 3

Failed

Enroll

Spring Boot

ID: 3

Code: CS301

Credits: 4

Pending

Enroll

Database Systems

ID: 4

Task 2: Attribute Directives — ngClass and ngStyle

Goal: Apply dynamic CSS classes and inline styles using attribute directives.

Student Course Portal

[Home](#)
[Courses](#)
[Profile](#)

Java Programming

ID: 1

Code: CS101

Credits: 4

Passed

Enroll | Show Details

Angular

ID: 2

Code: CS201

Credits: 3

Failed

Enroll | Hide Details

This course contains lectures, assignments and quizzes.

Spring Boot

ID: 3

Code: CS301

Credits: 4

Pending

Enroll | Show Details

Database Systems

Task 3: Custom Directive and Custom Pipe

Goal: Build a reusable attribute directive and a custom pipe for the portal.

TS course-card.ts U TS credit-label-pipe.ts U × <> home.html U ▶ 🔗 🗑

student-course-portal > src > app > pipes > TS credit-label-pipe.ts > CreditLabelPipe

```
1  import { Pipe, PipeTransform } from '@angular/core';
2
3  @Pipe({
4    name: 'creditLabel'
5  })
6  export class CreditLabelPipe implements PipeTransform {
7
8    transform(value: number | null): string {
9
10     if (value == null || value === 0) {
11       return 'No Credits';
12     }
13
14     if (value === 1) {
15       return '1 Credit';
16     }
17
18     return value + ' Credits';
19   }
20
21 }
```

Student Course Portal	
Home Courses Profile	
Java Programming	
ID: 1	
Code: CS101	
Credits: 1 Credit	
Passed	
Enroll	Show Details
Angular	
ID: 2	
Code: CS201	
Credits: 3 Credits	
Failed	
Enroll	Show Details
Spring Boot	
ID: 3	
Code: CS301	
Credits: No Credits	
Pending	
Enroll	Show Details
Database Systems	
ID: 4	
Code: CS401	